

4.1 Return on Investment (ROI)

Return on Investment (ROI) is a key financial metric used to evaluate the profitability or efficiency of an investment, expressed as a percentage.

It is calculated by dividing the net profit by the total cost of the investment and multiplying by 100, represented by the formula:

$$ROI = \left(\frac{\text{Net Profit}}{\text{Cost of Investment}} \right) \times 100.$$

A higher percentage indicates better returns, aiding in comparing different projects.

Key Aspects of ROI:

$$\frac{\text{Current Value of Investment} - \text{Cost of Investment}}{\text{Cost of Investment}} \times 100.$$

- **Formula:**
- **Purpose:** Measures the gain or loss generated relative to the amount of money invested.
- **Applications:** Used for evaluating stocks, marketing campaigns, real estate, and business ventures.
- **Limitations:** It does not account for the time period (holding period), meaning it may overlook opportunity costs.

Examples:

- **\$1,000 Investment → \$1,200 Return:** $\frac{(1200 - 1000)}{1000} \times 100 = 20\% \text{ ROI}.$
- **\$10,000 Investment → \$12,000 Return:** $\frac{(12000 - 10000)}{10000} \times 100 = 20\%$

Key Components:

- **Net Income/Profit:** The total gain from the investment minus all associated costs.
- **Cost of Investment:** The total expenses incurred, including initial outlay and maintenance.